

Final Project Report Template

Abstract

Write a concise summary of the project: problem, approach, prototype, results, and key takeaway.

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1 Introduction

1.1 Background & Motivation

Explain the context and the problem.

1.2 Project Goals

State the objectives of your project.

2 Project Description & Workflow

2.1 Problem Statement

Describe the problem you are addressing.

2.2 Overview of the Proposed Solution

Provide an overview of your solution and operating principles.

2.3 Workflow Diagram / System Flow

Insert a diagram showing data flow, sensors, control logic, etc.

Figure 1: System workflow diagram

3 Prototype Development — Hardware

Datasheet and description of the load of choice.

Describe hardware choices, ESP32 wiring, sensors used, rationale, and calibration steps.

Figure 2: Hardware configuration

Figure 3: Circuit diagram

4 Prototype Development — Software

External API? What are the data sources? etc.

Explain software architecture, data handling, control logic. Include block diagrams or pseudocode if it helps explaining.

Figure 4: Software architecture diagram

5 Testing and Validation Iterations

Testing/calibration of the sensors/hypothesis validation procedures.

6 Results

Present findings, performance metrics, savings, potential improvements, visualizations, etc.

Figure 5: Performance metrics

7 Conclusion

Summarize accomplishments, main takeaways, strengths, weaknesses.

8 References

List datasheets, tutorials, libraries, research papers, and other resources used.

References

[1] Author, “Title,” Journal, Year.

A Appendices

Include schematics, extended data, code snippets, etc.

A.1 Schematic Diagrams

A.2 Code Snippets